

8 Proofs Involving Sets

Use the methods introduced in this chapter to prove the following statements.

2. Prove that $\{6n : n \in \mathbb{Z}\} = \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$.

Proof. First we show $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. Suppose $a \in \{6n : n \in \mathbb{Z}\}$. This means $a = 6n$ for some integer n . Therefore $a = 2(3n)$ and $a = 3(2n)$. From $a = 2(3n)$, it follows that a is a multiple of 2, so $a \in \{2n : n \in \mathbb{Z}\}$. From $a = 3(2n)$, it follows that a is a multiple of 3, so $a \in \{3n : n \in \mathbb{Z}\}$. Thus by definition of intersection of two sets, we have $a \in \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. Therefore $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$.

Next we show $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$.

Suppose $a \in \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. By definition of intersection of two sets, this means $a \in \{2n : n \in \mathbb{Z}\}$ and $a \in \{3n : n \in \mathbb{Z}\}$. Since $a \in \{2n : n \in \mathbb{Z}\}$, we know $a = 2k$ for some $k \in \mathbb{Z}$. Thus a is even. Since $a \in \{3n : n \in \mathbb{Z}\}$, we know $a = 3l$ for some $l \in \mathbb{Z}$. Since 3 is odd, if l were odd then $3l$ would be odd; as $a = 3l$ is even, it follows that l is even. Then $l = 2j$ for some $j \in \mathbb{Z}$. So we have $a = 3l = 3(2j) = 6j$. Since $j \in \mathbb{Z}$, it follows that $a \in \{6n : n \in \mathbb{Z}\}$. Thus $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$.

Therefore, since $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$ and $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$, it follows that $\{6n : n \in \mathbb{Z}\} = \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. $\square$

4. If $m, n \in \mathbb{Z}$, then $\{x \in \mathbb{Z} : mn \mid x\} \subseteq \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$.

Proof. Suppose $a \in \{x \in \mathbb{Z} : mn \mid x\}$. Then $a = mnk$ for some $k \in \mathbb{Z}$. So we have $a = m(nk)$ and $a = n(mk)$. By definition of divisibility, $m \mid a$ and $n \mid a$. Thus $a \in \{x \in \mathbb{Z} : m \mid x\}$ and $a \in \{x \in \mathbb{Z} : n \mid x\}$. By definition of intersection of two sets, we have $a \in \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$. Therefore $\{x \in \mathbb{Z} : mn \mid x\} \subseteq \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$. $\square$

6. Suppose A, B and C are sets. Prove that if $A \subseteq B$, then $A - C \subseteq B - C$.

Proof. We prove the contrapositive: if $A - C \not\subseteq B - C$, then $A \not\subseteq B$. Assume $A - C \not\subseteq B - C$. Then there exists an element $x \in A - C$ such that $x \notin B - C$. Since $x \in A - C$, we have $x \in A$ and $x \notin C$. Since $x \notin B - C$, it follows that $x \notin B$ or $x \in C$. Because $x \notin C$, it follows $x \notin B$. Since $x \in A$ and $x \notin B$, by definition of a subset, we have $A \not\subseteq B$. $\square$